Sparse Koopman Autoencoders Identify Local Dynamical Regimes in Multibasin Systems

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Abstract

Koopman autoencoders (KAEs) seek a higher-dimensional latent representation in which nonlinear dynamics evolve linearly. However, many interesting systems have multiple basins of attraction, and both theoretical and empirical work has shown these multibasin systems cannot generally admit a single finite-dimensional global Koopman embedding under standard assumptions. We posit that encoders with a sparsity-inducing objective encouraging few active latent coefficients will provide latent supports as an inspectable basin-modeling principle for Koopman autoencoders. We use these encoders producing sparse latents in training Sparse Koopman Autoencoders (SKAEs) without basin labels or other regime annotations, and treat the learned latent supports as model-produced regime variables after training. Across a range of procedurally generated multibasin systems and chaotic flows, we show that SKAEs have superior forecasting performance compared to dense-latent KAEs. We also perform a mechanistic study that shows latent supports produced by SKAEs are both essential for the quality of the representation and useful for identifying basins on held-out basin interior states, whereas dense-latent KAEs collapse to an uninformative single family. These results identify sparse latents and their corresponding supports as label-free, interpretable regime variables for Koopman learning in nonlinear systems with multiple local dynamical laws.

1 Introduction

Learning representations for nonlinear dynamical systems remains a central problem in machine learning, scientific computing, and control. A particularly useful goal is to embed nonlinear dynamics into coordinates in which the evolution is linear, or at least locally linear, because linear models can be combined with mature tools for prediction, estimation, and control, including model predictive control and linear quadratic regulation [27, 15, 20, 17, 26]. This goal is also well motivated theoretically. Classical dynamical-systems results provide local topological linearizations near hyperbolic fixed points [14, 16], while Koopman eigenfunction constructions can yield linearizing coordinates on basins of attraction for stable equilibria and periodic orbits [22, 21]. Thus, nonlinear systems may admit linear dynamics with a specific change of coordinates.

The difficulty is that these results are primarily existential. They establish when useful coordinates can exist, but they do not by themselves provide a finite-data procedure for constructing such coordinates. Koopman operator theory formalizes the linearization idea by lifting nonlinear state evolution to a linear operator acting on observables [18, 19, 28, 6]. This viewpoint has inspired data-driven approximations of Koopman dynamics, such as DMD and eDMD [32, 35, 36], and more recently to Koopman autoencoders, which learn an encoder from state space to a latent representation, a linear latent transition matrix, and a decoder back to state space [34, 25, 30, 3, 33, 10]. These models provide an appealing route from existence results to practical learning: rather than hand-designing observables, they attempt to learn the embedding directly from trajectory data.

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A central challenge for Koopman approximations is that many practically interesting nonlinear systems typically contain multiple equilibria or basins of attraction; for example, a damped mechanical system can settle into different wells, a biological or chemical model can approach different stable equilibria, and a controlled system can move between regions with distinct local dynamics. However, theory shows that these systems cannot generally admit a single finite-dimensional global linearization with a continuous encoder-decoder pair [22, 5, 31, 24], and empirical evidence supports this in practice [10]. In such multistable systems, it is possible to linearize the dynamics within basins, but the linearizations needed in different basins of attraction are generally incompatible with each other. Therefore, for multi-attractor systems, the right objective is not to force all basins through one continuous global linearization, but to discover and use distinct local linearizations.

This paper proposes inducing sparse latent structure as a practical and interpretable method for Koopman learning of multi-attractor systems. By inducing the state to activate only a small subset of coordinates in lifted latent space, we can incentivize different regions of state to occupy different active coordinates, since nearby states should have similar latent embeddings, and thus enable local linearizations. In this sparse representation, nonzero coefficient values in a latent vector can provide continuous coordinates for prediction within a selected region, while the active indices provide a discrete support object that can be inspected across states and used to select local linearizations.

Sparse coding and LISTA-style encoders [13, 8, 1] provide the right inductive bias for this purpose. They naturally induce piecewise-linear, union-of-subspaces structure [29, 7, 9, 13], which is well-matched to this setting where different attractors or basins require different local linearizations in theory. We hypothesize that sparse-latent Koopman autoencoders will assign local dynamical regimes to distinct latent support families, yielding a learned regime variable without basin supervision. Then, periodically decoding and re-encoding the state can refresh the quality of the latent embedding over time by re-selecting the correct local dynamics when trajectories move across the state space [10].

Contributions. We introduce Sparse Koopman Autoencoders (SKAEs), showing that sparse latent supports can serve as label-free regime variables that reconcile basin-specific Koopman linearizations with the practicality of a single trained model. Across multibasin and chaotic benchmarks, SKAEs (i) improve long-horizon forecasting while (ii) producing supports that are functionally necessary for accurate rollouts and (iii) interpretable as emergent local dynamical regimes.

2 Problem formulation

The introduction motivates a multibasin version of Koopman learning. Given sampled trajectories of a nonlinear system, the aim is to learn a state-reconstructing latent representation whose evolution is linear, without being told which basin or local dynamical regime generated each observation. We formulate the problem at the cadence of the stored benchmark observations. For each system, let $\mathcal{X} \subseteq \mathbb{R}^{d_x}$ be the observed state space and let

$$x_{k+1} = F_{\Delta t}(x_k), \qquad x_k \in \mathcal{X}, \tag{1}$$

where $x_k \in \mathbb{R}^{d_x}$ is the state at the kth stored sample, $F_{\Delta t}$ denotes the transformation to the next discrete observation step based on underlying dynamics, and Δt is the system-specific observation interval. Forecasting is therefore defined by repeatedly applying the map $F_{\Delta t}$; a horizon of h means h applications of $F_{\Delta t}$; we denote k compositions as $F_{\Delta t}^{(k)}$. Because Δt may differ across systems, equal values of h may yield different physical time.

Basins of attraction. Many systems of interest are multistable: different initial conditions can approach different long-run behaviors. A fixed point is a state x^* satisfying $F_{\Delta t}(x^*) = x^*$. More generally, a trajectory may approach an attractor A, such as a stable equilibrium, limit cycle, countable finite set, or chaotic invariant set. The basin of attraction of A is the set of initial conditions whose trajectories converge to A:

$$\mathcal{B}(A) = \left\{ x_0 \in \mathbb{R}^{d_x} : \inf_{a \in A} \left\| F_{\Delta t}^{(k)}(x_0) - a \right\|_2 \to 0 \text{ as } k \to \infty \right\}. \tag{2}$$

When multiple attractors coexist, their basins partition the state space up to basin boundaries. This defines the setup for this paper. The task is to learn a representation that can support linear prediction where the appropriate linear law depends on where the state lies in the multibasin geometry.

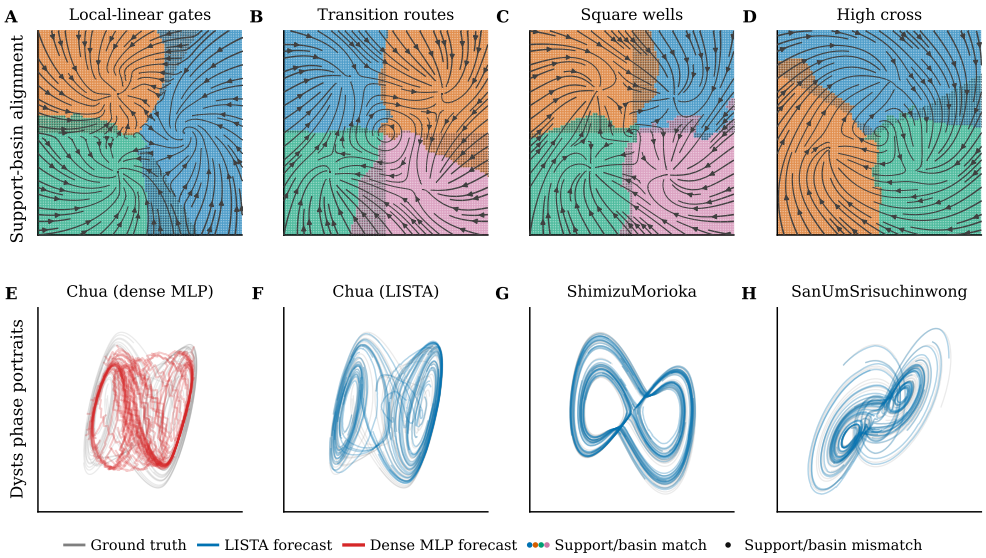


Figure 1: Top: procedurally generated multibasin systems where gray streamlines are the true vector field, colored dots mark support-family/basin matches, and black dots mark support-family/basin mismatches. The learned LISTA support families are mapped post hoc to each family's dominant evaluation basin. Bottom: Dysts phase portraits with held-out truth in gray and forecasts in color. Red lines are predictions by the seed-0 dense-latent MLP, and blue lines are from the best seed-0 LISTA variant. The dense MLP makes significant mistakes in forecasting the Chua system, unlike LISTA.

Koopman theory. The stored-step map $F_{\Delta t}$ induces a Koopman operator $\mathcal{K}$ on scalar observables $g : \mathcal{X} \to \mathbb{R}$ by composition, $(\mathcal{K}g)(x) = g(F_{\Delta t}(x))$. Although $F_{\Delta t}$ may be nonlinear, $\mathcal{K}$ is linear on the space of observables. A finite-dimensional Koopman approximation searches for a vector observable $\Phi : \mathcal{X} \to \mathbb{R}^{d_z}$ and a matrix K such that $\Phi(F_{\Delta t}(x)) \approx K\Phi(x)$, while retaining enough information to reconstruct the state.

Koopman autoencoders. We study Koopman autoencoders (KAEs) in this setting. The model learns an encoder $E_\theta : \mathbb{R}^{d_x} \to \mathbb{R}^{d_z}$, a decoder $D_\phi : \mathbb{R}^{d_z} \to \mathbb{R}^{d_x}$, and a latent linear transition matrix $K \in \mathbb{R}^{d_z \times d_z}$. For observation x_k at arbitrary timestep k, the corresponding open-loop h-step forecast is

$$\hat{x}_{k+h} = D_\phi\big(K^h E_\theta(x_k)\big)\,. \tag{3}$$

The same matrix K is applied at every step of the rollout, so any state-dependent structure required for prediction must be encoded in the learned representation rather than in a supplied regime label.

Training windows. Training uses windows of adjacent stored observations. For a batch of B windows of rollout length L, $\{x_{b,k}, x_{b,k+1}, \ldots, x_{b,k+L}\}_{b=1}^{B}$, where b indexes windows and ℓ indexes offsets within a window, the model encodes each observed state, rolls out from the initial encoded state, and decodes both reconstructed observations and latent forecasts. The encoded/reconstructed quantities are defined for $\ell = 0, \ldots, L$, while rollout/forecast quantities are defined for $\ell = 1, \ldots, L$:

$$z^{\mathrm{enc}}_{b,k+\ell} = E_\theta(x_{b,k+\ell}), \quad z^{\mathrm{roll}}_{b,k+\ell} = K^\ell z^{\mathrm{enc}}_{b,k}, \quad x^{\mathrm{rec}}_{b,k+\ell} = D_\phi(z^{\mathrm{enc}}_{b,k+\ell}), \quad \hat{x}_{b,k+\ell} = D_\phi(z^{\mathrm{roll}}_{b,k+\ell}). \tag{4}$$

The training losses described later are built from these reconstructed and rolled-out quantities. Each application of K is interpreted as one stored benchmark step, matching the observation map in eq. (1).

Model discovers basins unsupervised. In the intended training and deployment setting, the model is not given basin labels or trajectory-to-basin assignments. When benchmark basin labels are available, they are reserved for post-hoc evaluation; they are not used to choose the model, train the encoder or transition, or select supports for support-conditioned predictions.

3 Method

3.1 Sparse supports as local regime variables

A finite-dimensional Koopman representation for the multibasin systems in Section 2 should encode two complementary quantities: (i) continuous coordinates that linearize the dynamics within a basin, and (ii) indicate which local linearization is active. We use sparsity to couple these two roles. If the state x is represented by a sparse latent code z, then the support of z can act as a regime variable, while the nonzero coefficient values provide coordinates for the corresponding local linear representation.

Given an overcomplete dictionary D, sparse coding estimates z by solving the Lasso objective

$$z^\star(x) = \arg\min_{z \in \mathbb{R}^{d_z}} \frac{1}{2} \|x - Dz\|_2^2 + \lambda_{\text{sc}} \|z\|_1 , \qquad \lambda_{\text{sc}} > 0. \tag{5}$$

which finds a code z with few non-zero coefficients to reconstruct the input x. In other words, it encourages a sparse solution to the under-specified problem (classically, ℓ_0 instead of ℓ_1 will find the sparsest solution with the fewest non-zero coefficients).

The classical solver to this problem (ISTA [4]) can be unrolled into a depth-L network of ISTA iterations, yielding learned ISTA (LISTA) [13]; in our setting, LISTA is an encoder yielding the sparse code $z = E_\theta(x)$, where the *support* is the set of active z coordinates. With a linear decoder, the support determines which decoder columns participate in reconstructing x, and the associated nonzero coefficients specify coordinates within that selected reconstruction subspace. When D and z are learned jointly, as in sparse coding [29], this induces a data-adaptive partition of the input state space where each region can be locally reconstructed with a code having the same support.

We therefore posit that combining Koopman representations and sparse coding objectives in a sparse Koopman autoencoder (SKAE) implicitly captures multibasin attractor dynamics without explicit basin labels in training. The Koopman objective encourages the continuous coefficients on each support to evolve linearly, while the sparsity objective encourages different local regimes to use different active coordinates. Thus, the model can represent a multibasin system using a discrete support pattern to identify the active basin and continuous coefficients for its localized Koopman (linear) representation. When ground-truth basin labels are available in benchmarks, we use them only after training to assess support–basin alignment or to construct diagnostic interventions.

3.2 Koopman autoencoder and sparse encoder

The predictor follows the KAE formulation in Section 2: an encoder, linear decoder, and latent transition, all of which are optimized jointly, produce the rollout in eq. (3). Like in sparse coding, [13] we constrain the columns of the decoder $D_\phi = D$ to be unit norm to avoid degenerate solutions. We experiment with K that is dense, block diagonal, or softly block-regularized.

Sparse encoding. Our main sparse encoder is LISTA [13]. It first computes a dense pre-code $c_t = f_\theta(x_t)$, then applies learned shrinkage refinements for $q = 0, \ldots, Q-1$:

$$u_t^{(0)} = \text{shrink}(c_t, \tau), \qquad u_t^{(q+1)} = \text{shrink}(S u_t^{(q)} + c_t, \tau), \qquad z_t = u_t^{(Q)}. \tag{6}$$

Here, $\text{shrink}(a, \tau) = \text{sign}(a) \max(|a| - \tau, 0)$ is applied element-wise, S is learned, and τ controls the threshold scale for shrinkage. The thresholding step creates exact zeros, so the encoder explicitly selects active latent coordinates while learning the selection rule from the prediction objective.

Transition families. We vary the structure of K separately from the encoder. The dense transition leaves K unrestricted. The block-diagonal transition partitions latent coordinates into M fixed groups and constrains

$$K_{\text{block}} = \text{blkdiag}(K_1, \ldots, K_M). \tag{7}$$

We also use a soft-block LISTA ablation that keeps K dense but penalizes cross-group entries,

$$\mathcal{L}_{\text{offblock}} = \sum_{i,j:\, g(i) \neq g(j)} |K_{ij}|, \tag{8}$$

where $g(i)$ is the fixed architectural group containing coordinate i.

3.3 Training objective

Training uses adjacent observation windows. For a batch of B windows with prediction horizon L, the model produces reconstructed future states $x^{\mathrm{rec}}_{b,k+\ell}$, latent rollouts $z^{\mathrm{roll}}_{b,k+\ell}$, encoded future latents $z^{\mathrm{enc}}_{b,k+\ell}$, and decoded rollout predictions $\hat{x}_{b,k+\ell}$ as defined in eq. (4). The rollout is seeded by the initial state, and the following sums are over offsets $\ell = 0, \ldots, L$:

$$\bar{\mathcal{L}}_{\mathrm{pred}} = \frac{1}{BL\sqrt{d_x}} \sum_{b,\ell} \left\| \hat{x}_{b,k+\ell} - x_{b,k+\ell} \right\|_2, \qquad \bar{\mathcal{L}}_{\mathrm{rec}} = \frac{1}{BL\sqrt{d_x}} \sum_{b,\ell} \left\| x^{\mathrm{rec}}_{b,k+\ell} - x_{b,k+\ell} \right\|_2, \tag{9}$$

$$\bar{\mathcal{L}}_{\mathrm{lin}} = \frac{1}{BL} \sum_{b,\ell} \left\| z^{\mathrm{roll}}_{b,k+\ell} - z^{\mathrm{enc}}_{b,k+\ell} \right\|_2, \qquad \bar{\mathcal{L}}_{\mathrm{sp}} = \frac{1}{BL} \sum_{b,\ell} \left\| z^{\mathrm{roll}}_{b,k+\ell} \right\|_1. \tag{10}$$

The observation-space terms are normalized by $\sqrt{d_x}$ so that their scale is comparable across systems of different state dimension. The latent terms are left in their native scale because the latent dimension is fixed within each comparison.

The total objective is

$$\mathcal{L} = \left(\lambda_{\mathrm{pred}} \bar{\mathcal{L}}_{\mathrm{pred}} + \lambda_{\mathrm{rec}} \bar{\mathcal{L}}_{\mathrm{rec}} + \lambda_{\mathrm{lin}} \bar{\mathcal{L}}_{\mathrm{lin}} + \lambda_{\mathrm{sp}} \bar{\mathcal{L}}_{\mathrm{sp}} \right) + \mathcal{L}_{\mathrm{struct}}. \tag{11}$$

Here $\mathcal{L}_{\mathrm{struct}} = 0$ for the dense and block-diagonal transition variants. For the soft-block transition, $\mathcal{L}_{\mathrm{struct}} = \lambda_{\mathrm{off}} \mathcal{L}_{\mathrm{offblock}}$, which probes whether a soft transition grouping is sufficient to induce useful support structure. The prediction and latent-linearity terms make the representation useful for Koopman rollout, the reconstruction term keeps the code tied to the observed state, and the sparsity term encourages selective activation.

3.4 Support objects

In this subsection, we introduce several notions of support that are used at evaluation time.

Let $z = E_\theta(x) \in \mathbb{R}^{d_z}$. A *support rule* first converts the latent code z into a Boolean active-coordinate vector $m(x) = \{|z_i| > 0\} \in \{0,1\}^{d_z}$, where $m_i(x) = 1$ means that latent coordinate i is active for state x. Equivalently, the exact support is the index set $S(x) = \{i : m_i(x) = 1\}$. Exact supports defined by an arbitrary threshold can be unstable: a small perturbation in x-space can lead to a different support in z, thus a single basin may be covered by several nearby or partially overlapping active-coordinate vectors. We therefore distinguish exact supports from *support families*, which group nearby Boolean support vectors and test whether these groups form basin-scale objects. Support families in general are more robust and lead to less basin fragmentation.

We use three support constructions:

- **Absolute-threshold support.** The absolute support vector is the Boolean mask denoted $m_{\mathrm{abs}}(x)$, where $m_{\mathrm{abs},i}(x) = \mathbf{1}\{|z_i| > 10^{-3}\}$ for indices i. $S_{\mathrm{abs}}(x) = \{i : m_{\mathrm{abs},i}(x) = 1\}$ is the equivalent active-index set. $m_{\mathrm{abs}}(x)$ is the input to support families such as F_{abs} (defined next), and to canonical wrong-support interventions.
- **Absolute-threshold support family.** F_{abs} is a group label for exact supports; every exact support m_{abs} is assigned an F_{abs} label. F_{abs} groups exact supports whose active-index sets substantially overlap. We define the overlap score as the Jaccard index, or intersection-over-union, between the active coordinates of two Boolean support vectors $m_{\mathrm{abs}}(x)$. On an evaluation collection $\mathcal{X}$, the algorithm first counts all distinct Boolean vectors and then visits those distinct vectors from most to least frequent, following a leader-style threshold clustering rule. Each family f stores one fixed representative vector r_f: the exact support vector that created that family. For the next distinct vector m, the algorithm computes the Jaccard index

$$J(m, r_f) = \frac{\sum_i \mathbf{1}\{m_i = 1 \text{ and } r_{f,i} = 1\}}{\sum_i \mathbf{1}\{m_i = 1 \text{ or } r_{f,i} = 1\}},$$

  with $J(0,0) = 1$ when both vectors are all zero. If the best existing representative has Jaccard similarity at least $\tau = 0.5$, every occurrence of m_{abs} receives that family label. Otherwise, m_{abs} starts a new family and becomes the representative r_f for that family. Pseudocode is provided in Appendix A. We write $F_{\mathrm{abs}}(x)$ for the family assigned to $m_{\mathrm{abs}}(x)$.

3.5 Periodic re-encoding

If a support indicates the currently relevant local dynamics, a long rollout should be able to update that support. We therefore evaluate periodic re-encoding, following Fathi et al. [10]. Starting from $z_k = E_\theta(x_k)$ and a re-encoding period m, the model advances for m Koopman steps, decodes the predicted state, and encodes that prediction again:

$$\tilde{z}_{k+m} = K^m z_k, \qquad \tilde{x}_{k+m} = D_\phi(\tilde{z}_{k+m}), \qquad z_{k+m} = E_\theta(\tilde{x}_{k+m}). \tag{12}$$

The same procedure is repeated over the rollout horizon. Each rollout uses only the model's predicted state; it does not use the true future state, a basin label, or an oracle for transitions between basins. For each dynamical system, we tune a separate re-encoding period m on the training set.

4 Experiments

The experiments aim to evaluate if finite-dimensional Koopman representations can be learned for multi-attractor systems implicitly without resorting to side information such as basin labels. We first discuss the quality of learned Koopman representations from sparse KAEs via a forecasting-based evaluation. Next, in the context of sparse inference, we evaluate the importance of the specific active set of coordinates for forecasting within a basin of attraction. Finally, we evaluate alignment of support families F_{abs} with withheld basin labels.

Benchmark systems. The main benchmark for our experiments contains 15 procedurally generated two-dimensional multibasin systems. These systems are constructed by specifying potential wells around known attractor coordinates, and superimposing dynamics that encourage transitions to different far-away regimes; see Appendix D. Because we specify the locations of the potential wells, the basin labels are well-defined and used only for evaluation. We also use a subset of 10 chaotic systems from the Dysts repository [11, 12] with the step size dt multiplied by a factor of 30 as an additional forecasting test. Analytic expressions and other details are in Appendix D, E.

Training models. We compare six models shown in Table 1 by varying the encoder and latent transition structure and studying their effects in this setting. We test three LISTA sparse encoders (where LISTA-BD denotes block diagonal latent transitions and LISTA-SB denotes soft-block regularization), two sparse-latent MLP applying the sparsity penalty $\bar{\mathcal{L}}_{\text{sp}}$ (10) with dense [10] or block-diagonal (BD) transitions, and a dense-latent MLP control baseline with no sparsity incentive. All models use a linear decoder with columns having unit norm constraint.

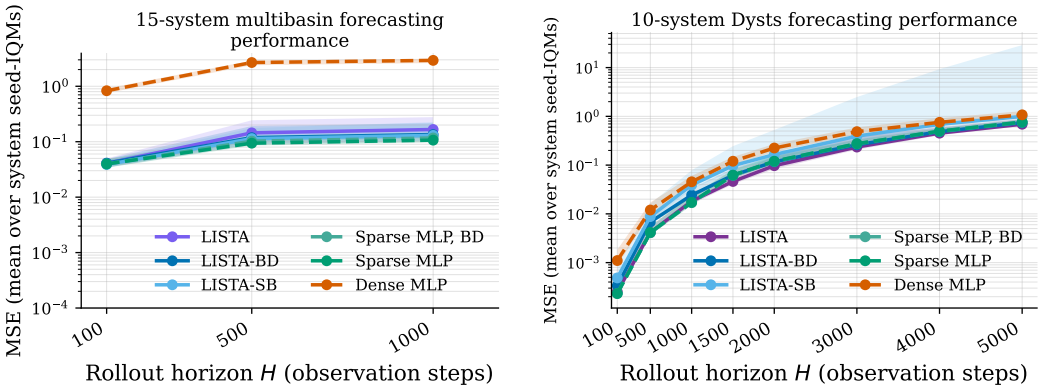
Most of these architectures are biased toward sparse inference by varying degrees. Our contribution is in studying how different ways of inducing sparsity can be useful. For LISTA, encoded states are sparse by construction via thresholding, and rollout latents get sparsity pressure from (10). For sparse MLPs, the encoded states have induced sparsity through the ReLU activations and ℓ_1 regularization [37, 23]. The dense MLP baseline has no intended zero-producing mechanism. Each model is trained for 15 random seeds on each system before evaluation. Training details are reported in Appendix B.

Statistical details. Point estimates are interquartile means (IQM) over seeds within each system to reduce seed uncertainty [2], then arithmetic means across systems. Horizon-trend lines use the same point estimate as the tables, with seed-bootstrap bands that summarize uncertainty over seeds for an average system. Statistical significance is rigorously tested with paired Wilcoxon signed-rank tests and Holm corrections; details are provided in Appendix C.

4.1 Forecasting experiments

Sparse encoders learn high-quality Koopman representations. The quality of the representation learned by a KAE is primarily measured by its ability to make accurate, multi-step forecasts. Thus, we evaluate the forecasting performance of trained sparse and dense-latent KAEs on the two benchmarks spanning 25 systems. Forecasting is evaluated on held-out trajectories at the stored observation cadence and summarized by mean squared error (MSE).

The results (Figure 2 and Table 1) show that sparse models yield a significant reduction of MSE forecasting error at all horizons on both sets of systems. In particular, at longer horizons, the dense MLP has 17-27x larger MSE at $H1000$ on the multibasin systems and about 1.5x larger MSE



(a) 15-system multibasin benchmark.

(b) 10-system Dysts $dt\times30$ benchmark.

Figure 2: Forecasting horizon trends; sparse models forecast better than dense models. Lines show arithmetic means across per-system seed-IQM MSEs on a log scale. Bands show fixed-system seed-bootstrap 95% intervals after system-wise log-relative normalization, anchored to the displayed mean, so they summarize seed uncertainty within an average system.

at $H4000$ on the Dysts systems compared to most of the sparse models (the LISTA-SB model outperforms the baseline but is noticeably worse than the rest). On the other hand, there are marginal differences between MSE predictions among the sparse-latent encoders. These results suggest that having encoded sparsity is crucial for good performance over all horizon prediction lengths.

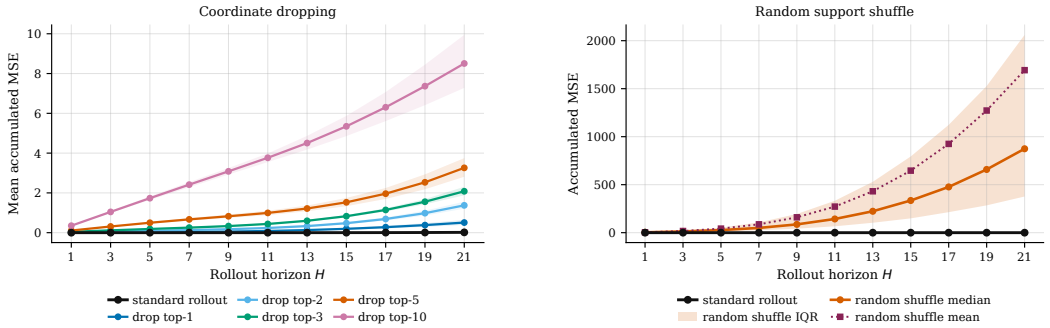
Sparse supports are essential for good representations. If a support formed by sparse encoders is meant to carry basin-relevant dynamical information, then changing only the initial active set should damage forecasts, even when the coefficient values are otherwise kept fixed. We do an ablation study to show whether accurate forecasts within a basin of attraction require the correct coordinates to be active by isolating the selection of sparse coordinates from their values. We do this by forcing a sparse KAE to use an incorrect active set with specific interventions, and evaluating 20-step forecasting performance under these settings. The interventions are the following:

1. **Coordinate dropping:** Force the coordinates of z with the $k \in \{1, \ldots, 10\}$ largest magnitude coefficients to zero before forecasting.
2. **Random support:** the active coefficient values are preserved, but reassigned to randomly chosen inactive latent coordinates.

The results show that these interventions have a significant detrimental effect on forecast quality, even on very short horizons. At $H = 21$, the standard rollout has mean accumulated MSE 0.0158, compared to 0.508/1.37/2.08/3.26/8.51 for the rollouts after dropping the top 1/2/3/5/10 active

Table 1: Forecasting performance and compact basin-support diagnostics. Forecasting values are MSEs; lower is better. Each cell is an arithmetic mean across systems after summarizing seeds within each system by IQM. Superscripts mark Holm-corrected tests against the dense-latent MLP baseline: multibasin forecasting cells use within-system Wilcoxon/Holm tests, Dysts cells use exact sign tests across systems, and multibasin support-diagnostic cells use per-basin deep-slice Wilcoxon/Holm tests ($*$, $p < 0.05$). **Bold** marks the best entry in each directional column.

	Forecasting MSE↓						Support diagnostics	
	Multibasin, 15 systems			Dysts $dt\times30$, 10 systems			Multibasin, 15 systems	
Model	H100	H500	H1000	H100	H2000	H4000	$H(B\,\vert\,F_{\text{abs}})\downarrow$	$\overline{\vert F_{\text{abs}}\vert}$
LISTA	0.0407^{*}	0.144^{*}	0.166^{*}	$2.53\times10^{-4*}$	$\mathbf{0.0972}^{*}$	$\mathbf{0.452}^{*}$	$\mathbf{0.130}^{*}$	**4.0**
LISTA-BD	0.0411^{*}	0.118^{*}	0.135^{*}	3.38×10^{-4}	0.117^{*}	0.485^{*}	0.156^{*}	3.8
LISTA-SB	$\mathbf{0.0387}^{*}$	0.114^{*}	0.130^{*}	4.85×10^{-4}	0.165	0.686	0.153^{*}	3.8
Sparse MLP, BD	0.0397^{*}	0.102^{*}	0.117^{*}	$2.33\times10^{-4*}$	0.119^{*}	0.493^{*}	0.257^{*}	3.3
Sparse MLP [10]	0.0397^{*}	$\mathbf{0.0940}^{*}$	$\mathbf{0.107}^{*}$	$\mathbf{2.32\times10^{-4*}}$	0.120^{*}	0.497^{*}	0.223^{*}	3.3
Dense MLP [25] *[baseline]*	0.830	2.68	2.93	0.00110	0.224	0.754	1.28	1.0



(a) Dropping the largest active coordinates.

(b) Moving active values to inactive coordinates.

Figure 3: Support-coordinate interventions on a multibasin system; interventions severely increase forecasting errors compared to the standard rollout. **(a)** The standard rollout is compared with cumulative dropping of the largest active coordinates of the initial sparse code; shaded bands are bootstrap 95% intervals over 100 initial conditions. **(b)** Active coefficient values are preserved but reassigned to randomly chosen inactive coordinates; the band is the interquartile range over shuffles.

Table 2: Accumulated MSE at $H = 21$ for the support interventions; lower is better.

Rollout	Mean ± SD	Median [IQR]
Standard	$\mathbf{0.0158 \pm 0.0127}$	$\mathbf{0.0140}$ $[\mathbf{0.0055}, \mathbf{0.0224}]$
Drop top-1	0.508 ± 0.647	0.218 $[0.148, 0.677]$
Drop top-2	1.37 ± 0.938	1.06 $[0.802, 1.63]$
Drop top-3	2.08 ± 1.10	1.80 $[1.34, 2.10]$
Drop top-5	3.26 ± 2.44	1.95 $[1.61, 4.17]$
Drop top-10	8.51 ± 6.83	5.16 $[4.58, 8.11]$
Random support	$1.69 \times 10^3 \pm 2.19 \times 10^3$	874 $[377, 2.06 \times 10^3]$

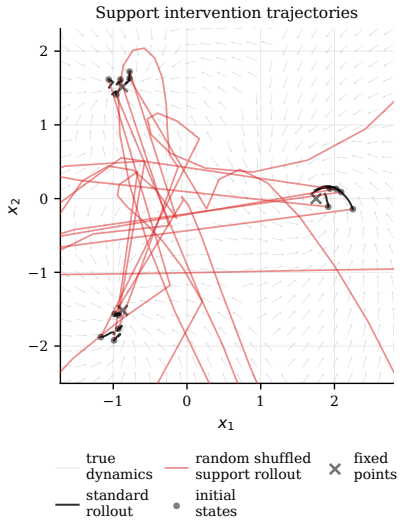
coordinates, and random support shuffling is much more destructive as seen in Figure 3. The trajectory panel (Figure 4a) shows the qualitative effects of the intervention. So, supports formed by sparse encoders do not merely correlate with basin labels; the active coordinates are key to the quality of the Koopman representation.

4.2 Support-basin alignment experiments

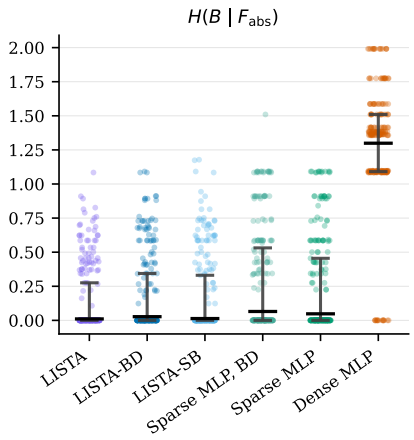
This mechanistic experiment examines the extent to which sparse support families fit after training can behave like basin-interior regime variables. States near separatrices are challenging and may confound the inferences: near these states, support changes can be induced by small perturbations in x, making supports unreliable for basin identification at separatrices. We therefore evaluate this diagnostic on held-out states deep in basins far from basin boundaries.

For a state x, let $d_1(x)$ and $d_2(x)$ be its distances to the nearest and second-nearest benchmark attractor centers. The *basin-depth margin* is $d_2(x) - d_1(x)$. This support-basin alignment experiment takes the top quartile of held-out states ranked by basin-depth margin within each benchmark basin, using ground-truth benchmark geometry to identify each basin. If a support family F_{abs} identifies a basin-specific regime used by the model, then on states deep inside a basin, it should leave little uncertainty about the withheld benchmark basin label B, measured by conditional entropy $H(B \mid F_{\text{abs}})$ as the main evaluation metric. $H(B \mid F_{\text{abs}})$ should be small when support families are basin-aligned. Also, different basins should be adequately represented by distinct support families, so we count the number of distinct support families formed in a system, denoted as $|F_{\text{abs}}|$, and compare it with the number of basins represented.

Support families F_{abs} identify basin membership. As seen in Table 1 and Figure 4b, LISTA F_{abs} support families provide more information to identify a ground truth basin label than any other model architecture, as seen by the lowest conditional entropy of 0.130. Other LISTA models with structured latent transitions are close competitors, followed by the sparse-latent MLPs, and then finally the dense-latent MLP baseline. LISTA-based models also expose the most support families, which better



(a) Randomly shuffled support trajectories (red) compared to standard rollouts (black). Randomly shuffling the active coordinates leads to divergences.



(b) Distribution of conditional entropy results; Dense MLP cannot identify basins based on support families. Dots show system–seed pairs, gray I-bars mark first-to-third quartile ranges, and black bars mark IQM summaries for $H(B \mid F_{\text{abs}})$.

matches the benchmark's basin-count mean/median 4.20/4; ideally, if a basin can truly be recovered by a single support family cluster, then the basin count should be one-to-one with the number of support families, $\overline{|F_{\text{abs}}|}$. In stark contrast, the dense-latent MLP collapses the representation to one support family, so distinct basins have correlated learned latent dynamics. This suggests that the support families induced by sparse coding may serve as a good proxy for ground-truth basin labels; the sparse LISTA models may have the emergent ability to discover basins unsupervised, despite this not being explicit in the training objectives.

5 Discussion

This paper shows that sparse Koopman autoencoders can learn useful local regime structure in multibasin dynamical systems without observing basin labels. Existing theory and empirical evidence indicate that multibasin systems generally should not be forced through one continuous finite-dimensional global Koopman embedding [22, 5, 31, 10, 24]. Our results show that sparse codes offer a practical alternative: a single trained model *can* find good representations for multiple local Koopman regimes by using different latent supports. This gives practitioners a simple diagnostic object to inspect after training.

We qualify some limitations. First, our interpretability claims are evaluated on benchmark systems where basin geometry is known, and the support–basin alignment analysis is intentionally restricted to states deep inside basins. Near separatrices, small state changes can imply very different long-term outcomes, and support assignments become fragmented or unstable. This is not a failure mode unique to sparse Koopman models, but it means that support families should be interpreted as reliable basin-interior regime variables rather than perfect global basin classifiers. Second, while the experiments include a diverse set of synthetic multibasin systems and external chaotic flows, they remain deterministic benchmarks. Real-world systems may contain observation noise, partial observability, nonstationarity, control inputs, higher-dimensional attractors, or regime changes that are not cleanly basin-like.

More broadly, this work argues that interpretability in dynamical representation learning can come from the structure of the latent code. Sparse supports provide an interface between continuous Koopman coordinates and discrete regime structure, making them a promising tool for multi-regime forecasting, analysis, and eventually control. By showing that these supports improve prediction, are necessary for accurate rollouts, and align with withheld basin information, this paper positions sparse latent structure as a practical step toward Koopman models for multibasin dynamics.

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A Support definitions and sensitivity

Section 3.4 defines the notation used in the main text. This appendix records the implementation-level objects behind that notation and explains how to read the corresponding diagnostics. All support objects are computed after training from the learned encoder output $z = E_\theta(x)$. Basin labels, basin counts, and trajectory-to-basin assignments are not used to train the model, define a support, or build a support family; when labels appear below they are evaluation-only quantities used to score or display the learned objects.

From a latent vector to an exact support key. The reducers first convert each latent vector into a Boolean mask $m(x) \in \{0,1\}^{d_z}$. The exact support $S(x)$ is the index set of the true entries of this mask, but the code compares exact supports by serializing the Boolean mask itself: two states have the same exact support if and only if all d_z Boolean entries match. The main support rule used in the paper is

$$m_{\mathrm{abs},i}(x) = \mathbf{1}\{|z_i| > 10^{-3}\}, \qquad S_{\mathrm{abs}}(x) = \{i : m_{\mathrm{abs},i}(x) = 1\}, \tag{13}$$

The inequalities are strict. S_{abs} is a magnitude-thresholded active set, so $|S_{\mathrm{abs}}|$ is a measured state-dependent statistic rather than a fixed sparsity budget.

From exact supports to support families. For a chosen support rule and evaluation collection, the implementation counts distinct exact support keys, sorts them from most to least frequent, and uses the serialized key as a deterministic tie-breaker. It then performs the leader-style greedy Jaccard merge described in Section 3.4. Each family stores an actual Boolean support vector as its representative. A new exact mask joins the existing representative with the largest Jaccard similarity if that similarity is at least 0.5; otherwise it starts a new family. Representatives are not optimized, averaged, or updated as centroids. Family labels are therefore run- and collection-specific integer identifiers, and only their induced partition of states is meaningful.

Support-family creation algorithm. For completeness, the family construction used for both F_{abs} and F_{top8} is:

> **Input:** evaluation states $\mathcal{X}$, a support rule q, and Jaccard threshold $\tau = 0.5$.
> **Output:** family label $F_q(x)$ for each $x \in \mathcal{X}$, and family representatives $\{r_f\}$.
>
> 1. Compute one exact Boolean mask $m_q(x) \in \{0,1\}^{d_z}$ for every $x \in \mathcal{X}$.
> 2. Count the multiplicity $c(m)$ of each distinct exact mask m.
> 3. Sort the distinct masks in decreasing $c(m)$, using the serialized Boolean mask as a deterministic tie-breaker.
> 4. Initialize an empty list of families and representatives.
> 5. For each distinct mask m in the sorted order:
>   - **a.** If no family exists, create a new family f, set $r_f \leftarrow m$, and assign m to f.
>   - **b.** Otherwise compute $f^\star = \arg\max_f J(m, r_f)$ over existing family representatives.
>   - **c.** If $J(m, r_{f^\star}) \geq \tau$, assign every occurrence of m to family $f^\star$.
>   - **d.** If $J(m, r_{f^\star}) < \tau$, create a new family f, set $r_f \leftarrow m$, and assign every occurrence of m to f.
> 6. For each state x, return $F_q(x)$, the family assigned to its exact mask $m_q(x)$.

The representative update $r_f \leftarrow m$ occurs only when a new family is created. Existing representatives are never averaged, recomputed as majority masks, or moved toward later assigned masks.

For top-eight masks, $J(m, r) \geq 0.5$ is equivalent to at least six shared active coordinates because both masks have size eight. For S_{abs}, the same threshold is adaptive to mask size through the intersection-over-union denominator. A lower Jaccard threshold merges more exact supports and can collapse distinct basins; a higher threshold separates more masks and can make $H(B \mid F)$ small by over-fragmenting each basin. We therefore fix 0.5 rather than tuning it post hoc and interpret any entropy result together with the corresponding family count.

Which support object is used where.

S_{abs}.
This is the strict active-coordinate object. It is used to build F_{abs}, to report exact-support diagnostics such as $H(B \mid S_{\mathrm{abs}})$, $H(S_{\mathrm{abs}} \mid B)$, exact-support uniqueness, and $|S_{\mathrm{abs}}|$, and to form canonical masks for wrong-support interventions. In the intervention code, the canonical mask for basin b is the most frequent S_{abs} key among candidate deep-slice states from basin b. The "wrong" condition replaces it with canonical masks from other represented basins and holds that mask fixed during the masked latent rollout.

F_{abs}.
This is the main basin-support alignment object in Table 1. It asks whether the many exact S_{abs} masks produced by a sparse encoder compress into basin-scale families. The primary directional metric is $H(B \mid F_{\mathrm{abs}})$: low values mean that knowing the support family leaves little uncertainty about the withheld basin label. The count $\overline{|F_{\mathrm{abs}}|}$ is deliberately non-directional and should be compared with the number of basin labels represented in the evaluated slice.

How to read the diagnostics. A low $H(B \mid S_{\mathrm{abs}})$ or $H(B \mid F_{\mathrm{abs}})$ means that the support object predicts basin identity on the evaluated benchmark states. It does not by itself imply a compact one-support-per-basin code: a model can achieve low $H(B \mid S_{\mathrm{abs}})$ while assigning many different exact masks to the same basin. $H(S_{\mathrm{abs}} \mid B)$, exact-support uniqueness, and $|S_{\mathrm{abs}}|$ diagnose this fragmentation, while $H(B \mid F_{\mathrm{abs}})$ and $\overline{|F_{\mathrm{abs}}|}$ ask whether the fragmented exact masks merge into a basin-scale family structure.

The takeaways are as follows. F_{abs} is the object for the claim that sparse supports identify basin interiors. S_{abs} is the intervention object for testing whether the active coordinates are functionally used.

B Additional experimental details

This appendix records the paper-facing experimental protocol. The main experiments ask whether sparse Koopman autoencoders learn useful latent supports without basin supervision. The model is

never given basin labels, basin counts, or trajectory-to-basin assignments when training the encoder, decoder, latent transition, support rules, or periodic rollouts. Benchmark basin metadata is used only after training to define evaluation slices, construct controlled diagnostic interventions, and score support–basin agreement. The only training-time exception is architectural: the block-diagonal and soft-block transition diagnostics use benchmark metadata to set the fixed number of transition groups, but they still do not receive trajectory labels or support labels.

Benchmarks. The controlled benchmark contains 15 two-dimensional multibasin systems, listed with their equations and attractor metadata in Section D. These systems are integrated to produce stored observation sequences; the learner observes only stored states, not vector fields, integration substeps, basin labels, or basin assignments. The external forecasting stress test uses the 10 three-dimensional Dysts $dt{\times}30$ systems listed in Section E. For Dysts, stored observations are generated at 30 times each system's native integration interval and coordinates are standardized after trajectory generation.

Model rows. The six model rows in Table 1 are LISTA, LISTA–BD, LISTA–SB, Sparse MLP, Sparse MLP–BD, and Dense MLP. All use latent dimension $d_z = 256$. LISTA rows use shrinkage to produce exact zeros; MLP sparse rows use the sparsity penalty in eq. (10); Dense MLP removes the intended sparsity mechanism and sets the sparsity coefficient to zero. The BD rows use a block-diagonal latent transition. The SB row keeps a dense transition but adds an off-block penalty. All rows use a linear decoder dictionary as the state decoder. The controlled multibasin LISTA rows use threshold scale $\alpha = 0.15$, two shrinkage-refinement loops, and a sign-split final operation. The Dysts $dt{\times}30$ LISTA rows use the same threshold scale but with one refinement loop and a ReLU final operation. MLP encoders use two hidden layers of width 64; Dense MLP uses a tanh encoder without the sparse output gate. LISTA uses MLPs of the same specification (depth and width) to produce the precode for refinement.

Controlled multibasin training. For every controlled system, each model row is trained for seeds $0, \ldots, 14$. Training uses rollout length $L = 8$ windows, meaning 9 adjacent stored states per window, for 200,000 optimization steps with minibatches of 256 windows. The optimizer is AdamW. Encoder and decoder parameters use learning rate $5{\times}10^{-5}$, the latent transition uses learning rate $5{\times}10^{-6}$, and non-transition parameters use weight decay 10^{-4}. In eq. (11), the loss weights are $\lambda_{\mathrm{pred}} = 1$, $\lambda_{\mathrm{rec}} = 0.03$, $\lambda_{\mathrm{lin}} = 1$, and $\lambda_{\mathrm{sp}} = 3{\times}10^{-3}$ for sparse rows; Dense MLP uses $\lambda_{\mathrm{sp}} = 0$. Soft-block rows add the off-block L_1 transition penalty with weight 10^{-4}.

All controlled rows use the same label-free boundary-emphasized reset distribution. Before training, 4096 candidate initial states are probed for 32 steps. Candidates are scored by short-horizon perturbation sensitivity and residual late-rollout motion, and the top 1024 states are retained. The scoring probe uses four perturbations, perturbation scale 0.04, a late-transient window of 8 steps, and transient weight 0.5. Half of training windows start from this retained pool with jitter scale 0.25. The retained pool is intended to expose boundary-adjacent and transient regions that would otherwise be rare under ordinary resets; it is built without basin labels.

Dysts $dt{\times}30$ training. Dysts rows are trained for the same seeds, $0, \ldots, 14$, with latent dimension $d_z = 256$, minibatches of 256, rollout length $L = 10$ windows, and 100,000 optimization steps. Dysts trajectories come from deterministic native caches with 200 cached trajectories, 30,000 stored observations per trajectory, and a 2,000-step warm-up. The first cached trajectory starts from the Dysts default initial condition; remaining trajectories perturb that state with Gaussian noise at scale 0.2 times the coordinate-wise Dysts standard deviation. Train, validation, and test caches use separate deterministic namespaces, so held-out test windows are disjoint from training windows and reusable across model seeds. LISTA rows use the controlled-benchmark learning rates. MLP rows use learning rates 10^{-4} for encoder/decoder parameters and 10^{-5} for the latent transition. Sparse Dysts rows use $\lambda_{\mathrm{sp}} = 6{\times}10^{-3}$, while Dense MLP uses $\lambda_{\mathrm{sp}} = 0$.

Checkpoint selection. Checkpoint selection is fixed before test evaluation. Every 500 optimization steps, and again at the final step, the current model is rolled out for 200 stored steps from 16 fixed held-out initial states using every-step re-encoding. The checkpoint with the lowest final validation error under this proxy is used for reported evaluations. Final tables aggregate completed seeds; they do not select the best seed.

Training recipe selection. Exploratory sweeps varied LISTA depth, shrinkage scale, sparsity coefficient, sign-split encodings, transition structure, and MLP sparsity controls. These sweeps used shorter budgets or broader system lists and are not pooled with the final evidence. The paper-facing protocol is the fixed recipe in this appendix: 15 seeds per retained system, the training budgets above, held-out evaluation splits, and the statistical units in Section C.

Forecasting experiments. Forecasting is evaluated on held-out trajectories at the stored observation cadence. A horizon h always means h stored observation steps, not a common physical time across systems. The no-reencoding rollout applies the learned latent transition for the full horizon before decoding. The periodic rollout advances the model's own predicted latent state, decodes the predicted state at a fixed period m, and re-encodes that predicted state; it never uses the true future state. Controlled multibasin forecasting uses 100 held-out initial conditions for each system and seed and reports the best score over the fixed period grid $m \in \{10, 25, 50, 100\}$. Dysts long-horizon forecasting uses 100 held-out test-cache windows for each system and seed and reports the best score over $m \in \{10, 25, 50, 100, 150, 200\}$. These grids are fixed before aggregation and are not tuned separately for each test trajectory.

Support–basin alignment. Support alignment is computed after training from held-out controlled-system states. For each state x, the basin-depth margin is the distance to the second-nearest benchmark attractor center minus the distance to the nearest attractor center. Within each represented benchmark basin, the evaluator keeps the top quartile of held-out states by this margin. This per-basin deep slice uses ground-truth geometry only to define the evaluation population. It is the clean slice for testing basin-support alignment because boundary states can legitimately have unstable or mixed supports.

On this slice, the evaluator encodes states and constructs $S_{\mathrm{abs}}(x) = \{i : |z_i| > 10^{-3}\}$. Exact Boolean masks are grouped into F_{abs} support families by the deterministic greedy Jaccard rule in Sections A and 3.4 with threshold 0.5. The main directional metric is $H(B \mid F_{\mathrm{abs}})$, where B is the withheld benchmark basin label; lower values mean that knowing the learned support family leaves less uncertainty about basin identity. The companion count $|F_{\mathrm{abs}}|$ is descriptive and is interpreted relative to the number of represented basins, not as a monotone score to maximize.

Support-coordinate intervention figures. The representative coordinate-intervention experiment isolates active-coordinate identity from coefficient values. Starting from a held-out basin-interior state, the model encodes $z_0 = E_\theta(x_0)$, perturbs only z_0, and then runs the ordinary learned transition and decoder without retraining. In the coordinate-dropping condition, the active entries are ranked by $|z_{0,i}|$ and the top $k \in \{1, 2, 3, 5, 10\}$ are set to zero before rollout. In the random-support condition, active coefficient values are preserved but reassigned to randomly chosen inactive coordinates. The figure and table use 100 held-out basin-interior starts; the random-support condition uses 20 random inactive-coordinate reassignments for each start. Errors are accumulated through $H = 21$, with the full horizon curves shown in Figure 3 and the $H = 21$ summary in Table 2.

Aggregation and significance. Point estimates in forecasting and support tables summarize seeds within each system by interquartile mean and then average system summaries across systems. Multibasin forecasting, support-alignment, and wrong-support significance annotations use paired seed-level effects within each controlled system against Dense MLP, with one-sided alternatives fixed by the table direction and Holm correction across eligible systems. Dysts forecasting uses each Dysts system as the independent unit after within-system seed-IQM summarization and applies a paired one-sided exact sign test with Holm correction. Support-refresh and coordinate-intervention panels are descriptive mechanism diagnostics. Full statistical definitions and denominator conventions are in Section C.

Hardware and resource details. Training was done on a SLURM cluster; RTX8000 was the main GPU being used. Training jobs load CUDA 12.6.0 and request one cluster GPU, 4 CPU cores, and 16 GB host memory per active training allocation. Controlled multibasin training uses single-run GPU allocations with a 24-hour time limit. Dysts $dt\times30$ training uses packed GPU allocations with up to 12 runs per allocation, one GPU, 4 CPU cores, 16 GB memory, and a 72-hour time limit per packed allocation. Dysts cache construction uses CPU allocations with 4 CPU cores, 16 GB memory, and a 24-hour time limit. Dysts long-horizon evaluation uses CPU allocations with 4 CPU cores, 24 GB memory, and a 6-hour time limit for the paper-facing $dt\times30$ evaluation queue. Controlled

support diagnostics and support-routed analyses use CPU allocations with 4 CPU cores, 16–24 GB memory, and 8–12-hour time limits, depending on the diagnostic.

C Statistical testing protocol

This appendix describes exactly what the significance annotations test. Point estimates and statistical annotations answer different questions. Point estimates in the forecasting and support-diagnostic tables first summarize completed training seeds within each benchmark system by an interquartile mean (IQM), then average those system summaries across systems. The main exception is the descriptive family-count column $\overline{|F_{\mathrm{abs}}|}$, which averages per-system seed means because it is a count diagnostic rather than a directional performance metric. Significance annotations instead ask whether a paired effect is reproducible under the independent unit implied by the experimental design. We therefore do not pool seeds, trajectories, transfers, and systems into one sample.

Common paired effects. For MSE comparisons against the dense-latent MLP baseline in Table 1, the paired effect for system s, replicate u, and horizon h is

$$\Delta_{s,u,h} = \log_{10} E^{\mathrm{cand}}_{s,u,h} - \log_{10} E^{\mathrm{Dense}}_{s,u,h} = \log_{10}\left(E^{\mathrm{cand}}_{s,u,h}/E^{\mathrm{Dense}}_{s,u,h}\right), \tag{14}$$

where E is the held-out best-periodic MSE and the one-sided alternative is $\Delta < 0$. Log ratios are used only for positive MSE ratios. They match the multiplicative scientific question and reduce the influence of the heavy-tailed seed-level MSE scale.

Support diagnostics use raw paired differences unless the displayed quantity is already an MSE ratio. Mean $|S_{\mathrm{abs}}|$ and $H(B \mid F_{\mathrm{abs}})$ are tested against Dense MLP with alternative candidate $<$ baseline.The family-count diagnostic $\overline{|F_{\mathrm{abs}}|}$ is not assigned a one-sided test because closeness to the represented basin count is not monotone in either direction.

Holm correction. All Wilcoxon families use Holm step-down correction at $\alpha = 0.05$. If $p_{(1)} \leq \cdots \leq p_{(m)}$ are the eligible raw p-values in a test family, the adjusted value at rank i is the running maximum of $\min\{1, (m-j+1)p_{(j)}\}$ for $j \leq i$, then mapped back to the original cell order. The implementation uses SciPy's one-sided Wilcoxon signed-rank test with Wilcox zero method; all-zero or otherwise degenerate paired differences are not counted as Holm passes.

Controlled multibasin forecasting and support diagnostics. For the controlled multibasin columns of Table 1 and the support-diagnostic columns of Table 1, the replicate unit is the training seed. Candidate and baseline values are paired by the same benchmark system and seed. For each candidate–metric–horizon–subset cell, the table-generating scripts run a separate one-sided paired Wilcoxon signed-rank test within each system over the finite paired seed deltas. MSE tests require positive finite MSEs before the log transform. The paper-facing table builders require at least four finite paired seeds for a system to be eligible for these within-system tests.

The raw per-system p-values are Holm-corrected across eligible systems for that cell. A compact superscript $*$ means the Holm-corrected test passes at 0.05 in the prespecified direction. The multibasin forecasting cells in the compact main table suppress the displayed K/N because every shown non-Dense multibasin forecasting cell passes on all $15/15$ systems.

Dysts forecasting superscripts. The Dysts columns in Table 1 use benchmark systems, not seeds, as the independent inferential units. For each Dysts system and horizon, seeds are first summarized within system:

$$I^{\mathrm{cand}}_{s,h} = \mathrm{IQM}_r\left\{E^{\mathrm{cand}}_{s,r,h}\right\}, \qquad I^{\mathrm{Dense}}_{s,h} = \mathrm{IQM}_r\left\{E^{\mathrm{Dense}}_{s,r,h}\right\}. \tag{15}$$

Each system then contributes one paired log effect

$$d_{s,h} = \log_{10} I^{\mathrm{cand}}_{s,h} - \log_{10} I^{\mathrm{Dense}}_{s,h}. \tag{16}$$

The compact Dysts table superscripts come from an exact one-sided sign test on the number of systems with $d_{s,h} < 0$. Equivalently, for n retained systems and w systems improved over Dense, the raw p-value is

$$\Pr\{\mathrm{Binomial}(n, 1/2) \geq w\}.$$

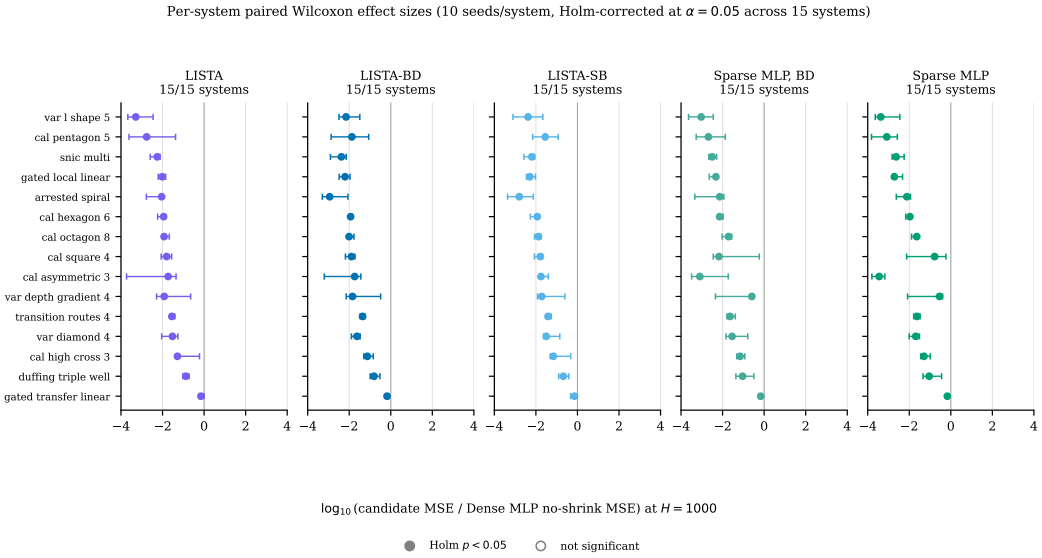


Figure 5: 15 Multibasin systems: Per-system effect sizes at $H{=}1000$ versus the Dense MLP baseline. Each point is one benchmark system; the x-coordinate is the per-system median paired $\log_{10}$-MSE difference (candidate minus baseline) over the common completed seeds, up to 15 per system, with whiskers showing the 95% bootstrap interval. Filled markers clear Holm-corrected $\alpha{=}0.05$.

These sign-test p-values are Holm-corrected across the non-Dense model–horizon comparisons in the Dysts analysis family, and the compact table reads the corrected sign-test values. The signed-rank and t-test values kept in the Dysts sidecar CSVs are audit diagnostics only; they do not determine the compact table superscripts. With 10 Dysts systems and this correction, a displayed Dysts superscript corresponds to improvement on all $10/10$ systems in that cell.

Support-coordinate interventions. Figure 3 is also a mechanism diagnostic rather than a cross-system significance table. The coordinate-dropping panel reports mean MSE curves over the selected held-out starts, with 95% bootstrap intervals from 2000 bootstrap resamples of starts. The random support-shuffle panel preserves active coefficient values, moves them to random inactive coordinates, repeats the shuffle procedure, and displays the interquartile range, median, and mean over the shuffle outcomes. These intervals describe intervention variability for the representative system and seed; they are not Holm-corrected hypothesis-test annotations.

D Multibasin benchmark inventory and system definitions

Table 3 lists the 15 two-dimensional multibasin systems used in this paper. Every row contributes to the retained-system aggregate in Table 1, and Figure 2a. Four systems are additionally displayed as the main benchmark visual panels in Figure 1. Claude Opus 4.5 in Claude Code provided substantial help in procedurally generating candidates given initial specifications, from which these 15 systems were chosen.

Basin labels are used only for benchmark annotation, controlled-transfer construction, and evaluation. Basin counts are benchmark metadata; for the structured-transition ablation rows, they are also used only to set a fixed architecture group count, as described in Section B, and not for support extraction, support-family construction, routing, or rollout evaluation.

Stored-step convention. All retained systems are implemented as continuous-time vector fields $\dot{x} = f_s(x)$ and integrated with fourth-order Runge–Kutta to produce the stored observation map $F_{\Delta t}$ used in eq. (1). The training windows contain only stored states x_k, not f_s, integration substeps, or basin assignments.

Table 3: The 15 multibasin benchmark systems.

Display name	System key	Generator family	Basins
Local-linear gates	`gated_local_linear`	piecewise local-linear gates	3
Transfer-gated local-linear	`gated_transfer_linear`	piecewise transfer gates	3
Arrested spiral	`arrested_spiral`	spiral capture with Gaussian traps	5
Asymmetric three-well	`cal_asymmetric_3`	Gaussian wells with independent rotation	3
High-cross three-well	`cal_high_cross_3`	Gaussian wells with stronger rotation	3
Hexagonal six-well	`cal_hexagon_6`	Gaussian wells with independent rotation	6
Octagonal eight-well	`cal_octagon_8`	Gaussian wells with independent rotation	8
Pentagonal five-well	`cal_pentagon_5`	Gaussian wells with independent rotation	5
Square four-well	`cal_square_4`	Gaussian wells with independent rotation	4
Triple-well Duffing	`duffing_triple_well`	triple-well Duffing oscillator	3
SNIC multi-attractor	`snic_multi`	polar SNIC-like phase dynamics	3
Transition-routes four-well	`transition_routes_4`	route-augmented Gaussian wells	4
Depth-gradient four-well	`var_depth_gradient_4`	Gaussian wells with depth gradient	4
Diamond four-well	`var_diamond_4`	Gaussian wells on a diamond layout	4
L-shaped five-well	`var_l_shape_5`	Gaussian wells on an L-shaped layout	5

Table 4: Parameters for retained Gaussian-well systems. Repeated pairs in the (a_i, σ_i) column apply to every listed center.

System	$\{c_i\}$	(a_i, σ_i)	(ω, γ)
High-cross three-well	$\{p_i^{(3)}(1.8, \pi/2)\}$	$(3.0, 0.5)$	$(2.0, 0.03)$
Hexagonal six-well	$\{p_i^{(6)}(1.7, 0)\}$	$(2.0, 0.5)$	$(1.0, 0.03)$
Octagonal eight-well	$\{p_i^{(8)}(2.2, 0)\}$	$(3.0, 0.5)$	$(0.90, 0.02)$
Pentagonal five-well	$\{p_i^{(5)}(1.8, \pi/2)\}$	$(2.0, 0.5)$	$(1.1, 0.03)$
Square four-well	$\{p_i^{(4)}(1.8, \pi/4)\}$	$(3.0, 0.5)$	$(1.0, 0.03)$
Diamond four-well	$\{(0, 2.2), (2.2, 0), (0, -2.2), (-2.2, 0)\}$	$(2.5, 0.5)$	$(1.0, 0.02)$
L-shaped five-well	$\{(-1.5, 1.5), (-1.5, 0), (-1.5, -1.5), (0, -1.5), (1.5, -1.5)\}$	$(2.5, 0.5)$	$(1.0, 0.03)$
Asymmetric three-well	$\{(0, 1.8), (-1.5, -0.9), (1.5, -0.9)\}$	$\{(2.5, 0.55), (1.5, 0.4), (2.0, 0.5)\}$	$(1.0, 0.03)$
Depth-gradient four-well	$\{(-1.3, 1.3), (1.3, 1.3), (1.3, -1.3), (-1.3, -1.3)\}$	$\{(2.2, 0.55), (2.5, 0.5), (3.0, 0.5), (3.5, 0.5)\}$	$(1.3, 0.03)$

Ground-truth vector-field visualizations. Figure 6 shows the continuous-time ground-truth vector fields for all 15 retained two-dimensional multibasin systems. Streamlines show the direction of $f_s(x)$, and black crosses mark the generator attractor centers or analytic equilibrium references used for benchmark annotation. The streamline colors use a panel-local $\log_{10} \|f_s(x)\|_2$ scale to reveal structure within each system; they should not be read as comparable speed scales across panels. Detailed panels with per-system colorbars are provided in Figures 7 to 11.

Gaussian-well family. Most retained catalog systems use a common two-dimensional Gaussian potential with an independent rotational drift. For $x = (x_1, x_2) \in \mathbb{R}^2$, let $Rx = (x_2, -x_1)$ and

$$V_s(x) = \sum_{i=1}^{M_s} -a_{s,i} \exp\left(-\frac{\|x - c_{s,i}\|_2^2}{2\sigma_{s,i}^2}\right) + \gamma_s(x_1^4 + x_2^4).$$

The implemented vector field is

$$\dot{x} = -\nabla V_s(x) + \omega_s Rx. \tag{17}$$

Define $p_i^{(M)}(r, \varphi) = r(\cos(2\pi i/M + \varphi), \sin(2\pi i/M + \varphi))$, $i = 0, \ldots, M-1$. The retained Gaussian-well systems instantiate eq. (17) with the parameters in Table 4.

The transition-routes system uses the same Gaussian-well form with centers

$$\{c_i\} = \{(-1.8, 1.8), (1.8, 1.8), (-1.8, -1.8), (1.8, -1.8)\},$$

$a_i = 3.0$, $\sigma_i = 0.6$, $\omega = 1.0$, and $\gamma = 0.03$, and adds a corridor-dependent rotation term:

$$\dot{x} = -\nabla V_s(x) + \big(1.0 + 0.3\, C_{\text{route}}(x_2)\big) Rx, \qquad C_{\text{route}}(x_2) = e^{-(x_2 - 1.8)^2/0.3} + e^{-(x_2 + 1.8)^2/0.3}. \tag{18}$$

Native gated local-linear systems. Both native gated systems place three centers on a circular layout, but with different radii and region definitions. Let $Q(\alpha) = \left(\begin{smallmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{smallmatrix}\right)$ and $\alpha_b = 2\pi b/3$, $b \in \{0, 1, 2\}$.

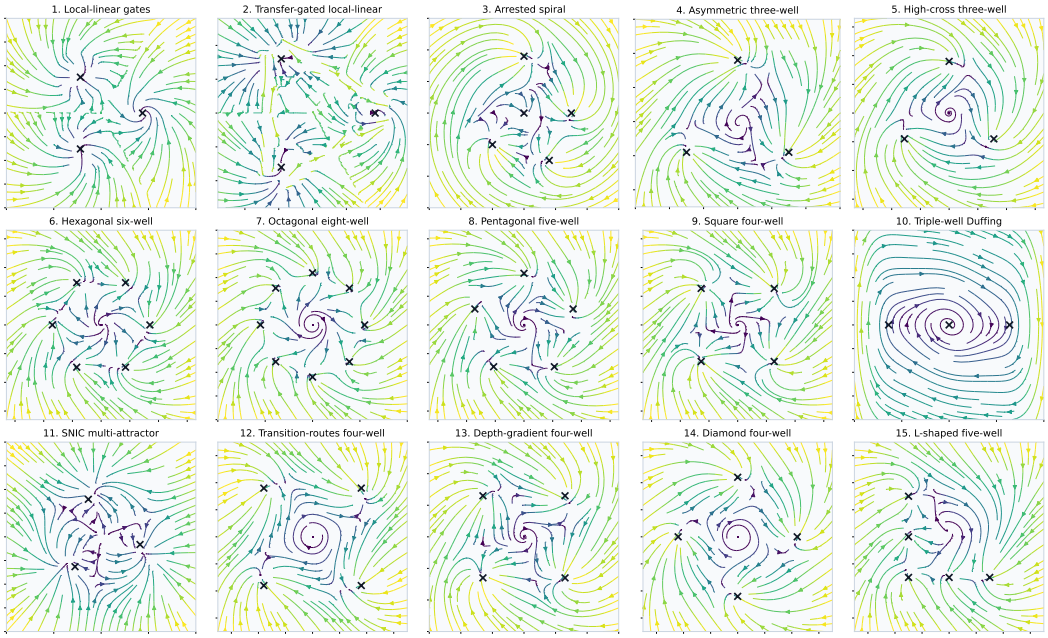
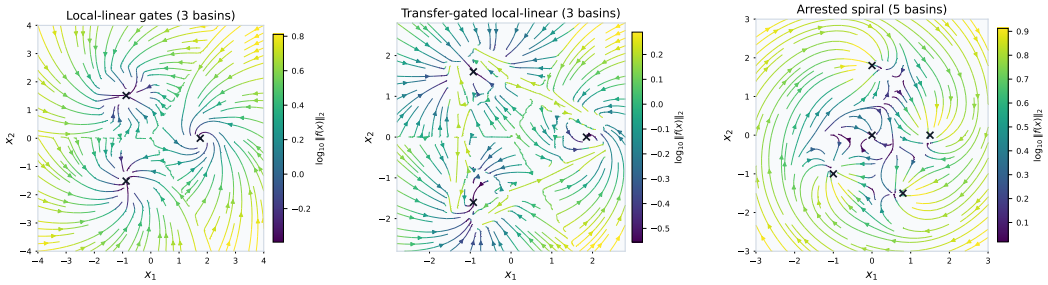
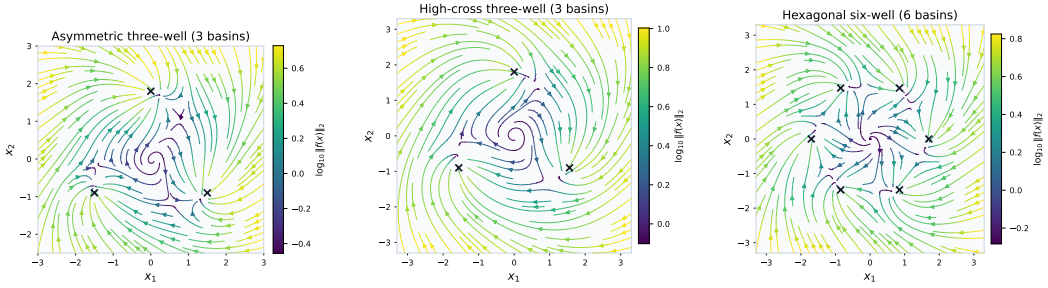


Figure 6: Ground-truth vector fields for the retained 15-system multibasin benchmark. The models are trained only on stored state trajectories; these continuous-time vector fields and attractor annotations are shown here only to document the benchmark geometry and are not provided to the learner.



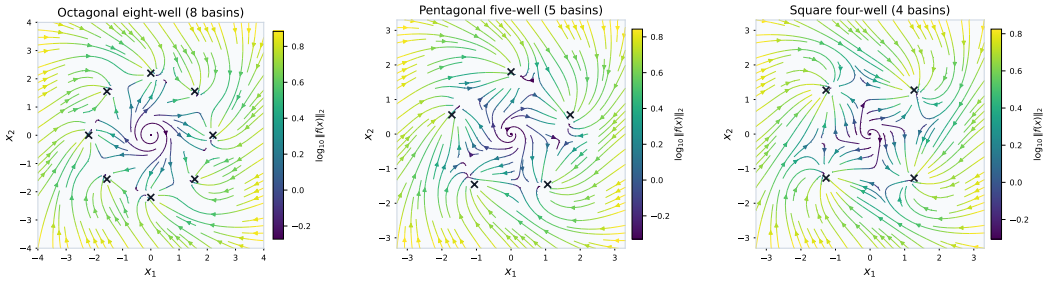
(a) Local-linear gates. (b) Transfer-gated local-linear. (c) Arrested spiral.

Figure 7: Detailed ground-truth vector-field panels for the first three retained multibasin systems. Colorbars report the panel-local $\log_{10} \|f_s(x)\|_2$ scale.



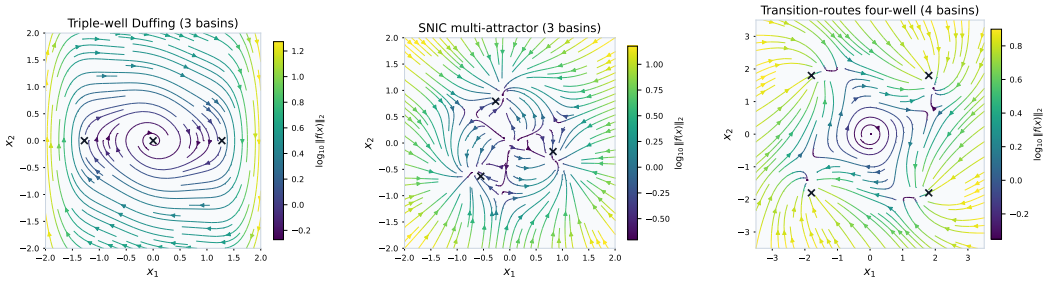
(a) Asymmetric three-well. (b) High-cross three-well. (c) Hexagonal six-well.

Figure 8: Detailed ground-truth vector-field panels for retained Gaussian-well systems. Colorbars report the panel-local $\log_{10} \|f_s(x)\|_2$ scale.



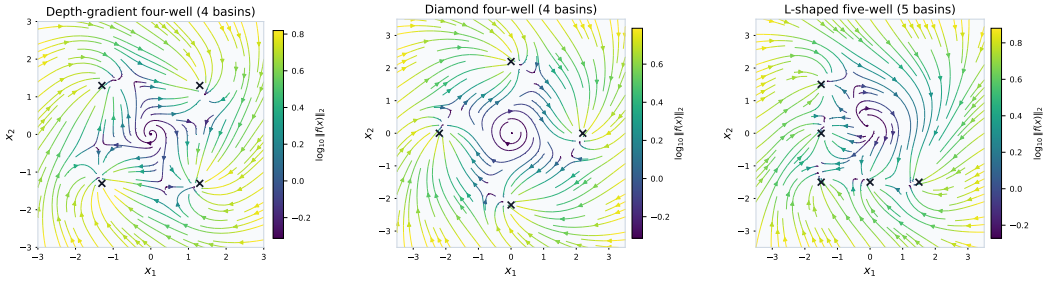
(a) Octagonal eight-well. (b) Pentagonal five-well. (c) Square four-well.

Figure 9: Detailed ground-truth vector-field panels for retained polygonal Gaussian-well systems. Colorbars report the panel-local $\log_{10} \|f_s(x)\|_2$ scale.



(a) Triple-well Duffing. (b) SNIC multi-attractor. (c) Transition-routes four-well.

Figure 10: Detailed ground-truth vector-field panels for specialized retained systems. Colorbars report the panel-local $\log_{10} \|f_s(x)\|_2$ scale.



(a) Depth-gradient four-well. (b) Diamond four-well. (c) L-shaped five-well.

Figure 11: Detailed ground-truth vector-field panels for retained geometry variants. Colorbars report the panel-local $\log_{10} \|f_s(x)\|_2$ scale.

For `gated_local_linear`, $c_b = 1.75(\cos\alpha_b, \sin\alpha_b)$. Inside the radius-1.05 core of basin b,

$$\dot{x} = A_b(x - c_b), \qquad A_b = Q(\alpha_b)\widetilde{A}_b Q(\alpha_b)^\top,$$

with

$$\widetilde{A}_0 = \left(\begin{smallmatrix} -0.9 & -1.2 \\ 1.2 & -0.9 \end{smallmatrix}\right), \quad \widetilde{A}_1 = \left(\begin{smallmatrix} -1.35 & 0.2 \\ -0.3 & -0.7 \end{smallmatrix}\right), \quad \widetilde{A}_2 = \left(\begin{smallmatrix} -0.7 & -0.1 \\ 0.5 & -1.2 \end{smallmatrix}\right).$$

Outside the cores, the active sector is the nearest angular sector $s(x)$ and the shared gate matrix

$$G = \left(\begin{smallmatrix} -1.35 & -0.9 \\ 0.9 & -1.35 \end{smallmatrix}\right)$$

drives the state as $\dot{x} = G(x - c_{s(x)})$.

For `gated_transfer_linear`, $c_b = 1.85(\cos\alpha_b, \sin\alpha_b)$. The implemented radii and widths are

$$\begin{aligned} \rho_{\text{core}} &= 0.30, \quad \rho_{\text{source}} = 0.80, \quad \rho_{\text{exit}} = 0.60, \quad \beta_{\text{exit}} = 0.72, \\ w_{\text{chan}} &= 0.22, \quad \delta_{\text{lane}} = 0.28, \quad \rho_{\text{hand}} = 0.45. \end{aligned}$$

The core matrices use the same rotated form with templates

$$\left(\begin{smallmatrix} -1.0 & -1.1 \\ 1.1 & -1.0 \end{smallmatrix}\right), \quad \left(\begin{smallmatrix} -1.4 & 0.2 \\ -0.2 & -0.7 \end{smallmatrix}\right), \quad \left(\begin{smallmatrix} -0.8 & -0.3 \\ 0.5 & -1.3 \end{smallmatrix}\right).$$

For each ordered pair $p = (s, t)$, $s \neq t$, define

$$d_{s\to t} = \frac{c_t - c_s}{\|c_t - c_s\|_2}, \qquad n_{s\to t} = (-d_{s\to t,2}, d_{s\to t,1}), \qquad o_t = \frac{c_t}{\|c_t\|_2},$$

and $\chi_{s,t} = 1$ when $\sin(\alpha_s - \alpha_t) \geq 0$, otherwise $\chi_{s,t} = -1$. The implemented channel entry and handoff points are

$$\begin{aligned} e_p &= c_s + \rho_{\text{source}} d_{s\to t} + \chi_{s,t}\delta_{\text{lane}} n_{s\to t}, \\ h_p &= c_t + \rho_{\text{hand}} o_t + \chi_{s,t}\delta_{\text{lane}}(-o_{t,2}, o_{t,1}), \end{aligned}$$

with channel direction $d_p = (h_p - e_p)/\|h_p - e_p\|_2$, channel normal $n_p = (-d_{p,2}, d_{p,1})$, and channel length $\ell_p = (h_p - e_p) \cdot d_p$.

The vector field is piecewise analytic. Core regions $\|x - c_b\|_2 \leq \rho_{\text{core}}$ use $A_b(x - c_b)$. Source annuli use $0.65A_b(x - c_b)$, and other non-channel background regions use $0.50A_b(x - c_b)$ for the nearest center c_b. Exit wedges for $p = (s, t)$ are source-neighborhood states outside the core with $\|x - c_s\|_2 \geq \rho_{\text{exit}}$ and $(x - c_s) \cdot d_{s\to t} \geq \|x - c_s\|_2 \cos\beta_{\text{exit}}$; they use

$$\dot{x} = v_{\text{exit}} d_{s\to t} - \lambda_{\text{exit}}\big((x - c_s) \cdot n_{s\to t}\big) n_{s\to t}, \qquad v_{\text{exit}} = 1.0, \quad \lambda_{\text{exit}} = 1.8;$$

and channel rectangles satisfying

$$0 \leq (x - e_p) \cdot d_p \leq \ell_p, \qquad |(x - e_p) \cdot n_p| \leq w_{\text{chan}}$$

use

$$\dot{x} = v_{\text{chan}} d_p - \lambda_{\text{chan}}\big((x - e_p) \cdot n_p\big) n_p, \qquad v_{\text{chan}} = 1.55, \quad \lambda_{\text{chan}} = 2.8.$$

Other specialized systems. The arrested spiral system has a background spiral flow plus four Gaussian traps and an origin trap:

$$\dot{x} = -0.3x + 2.0Rx - 4.0\sum_{i=1}^{4} e^{-\|x-c_i\|_2^2/(2\cdot 0.25)} \frac{x - c_i}{0.25} - 1.5e^{-\|x\|_2^2/0.3} \frac{x}{0.15} - 0.02(x_1^3, x_2^3),$$

with $c_i \in \{(1.5, 0), (0, 1.8), (-1, -1), (0.8, -1.5)\}$. Its fifth basin is the origin trap, matching the benchmark manifest count.

The triple-well Duffing system uses $x = (q, p)$ and

$$V(q) = q^6/6 - q^4/2 + aq^2, \qquad V'(q) = q^5 - 2q^3 + 2aq,$$

with $a = 0.3$, damping $\delta = 0.5$, rotation strength $\omega = 1.0$, and soft cubic confinement $-\epsilon u^3/s^3$ with $\epsilon = 0.003$ and $s = 4.0$:

$$\begin{aligned} \dot{q} &= (1 + \omega)p - \epsilon q^3/s^3, \\ \dot{p} &= -V'(q) - \delta p - \omega q - \epsilon p^3/s^3. \end{aligned} \tag{19}$$

Table 5: The ten Dysts systems used in the $dt\times30$ forecasting benchmark. The "stored step" column is 30 times the native Dysts integration interval.

Display name	System key	Dimension	Native dt	Stored step $30dt$
Chua	`Chua`	3	$2.8474745791\times10^{-4}$	$8.5424237373\times10^{-3}$
Dadras	`Dadras`	3	$6.5782963827\times10^{-4}$	$1.9734889148\times10^{-2}$
Dequan Li	`DequanLi`	3	$1.6763993999\times10^{-5}$	$5.0291981997\times10^{-4}$
Hadley	`Hadley`	3	$2.9086847808\times10^{-4}$	$8.7260543424\times10^{-3}$
Lu–Chen–Cheng	`LuChenCheng`	3	$1.8469678280\times10^{-4}$	$5.5409034839\times10^{-3}$
Qi–Chen	`QiChen`	3	$7.8371061844\times10^{-5}$	$2.3511318553\times10^{-3}$
Sakarya	`Sakarya`	3	$9.9704617436\times10^{-4}$	$2.9911385231\times10^{-2}$
San–Um–Srisuchinwong	`SanUmSrisuchinwong`	3	$1.4933288881\times10^{-3}$	$4.4799866644\times10^{-2}$
Shimizu–Morioka	`ShimizuMorioka`	3	$2.4080013336\times10^{-3}$	$7.2240040007\times10^{-2}$
Wang–Sun	`WangSun`	3	$5.3924987498\times10^{-3}$	$1.6177496249\times10^{-1}$

The SNIC system is implemented in polar form and then converted to Cartesian coordinates. With $\varepsilon_{\mathrm{num}} = 10^{-8}$, $r^2 = x_1^2 + x_2^2 + \varepsilon_{\mathrm{num}}$, $r = \sqrt{r^2}$, $\theta = \operatorname{atan2}(x_2, x_1)$, $c_\theta = x_1/(r + \varepsilon_{\mathrm{num}})$, and $s_\theta = x_2/(r + \varepsilon_{\mathrm{num}})$,

$$\dot{r} = r(1 - r^2) - 0.3\cos(3\theta), \qquad \dot{\theta} = 1 - 1.2\cos(3\theta).$$

The Cartesian vector field is

$$\begin{aligned}\dot{x}_1 &= \dot{r}c_\theta - r\dot{\theta}s_\theta - 0.01x_1(x_1^2 + x_2^2) + 0.5x_2,\\ \dot{x}_2 &= \dot{r}s_\theta + r\dot{\theta}c_\theta - 0.01x_2(x_1^2 + x_2^2) - 0.5x_1.\end{aligned}$$

These analytic generators define the benchmark trajectories. Held-out evaluation annotations are produced by the benchmark label helpers or by endpoint-rollout basin labeling, and the learned models observe only stored states.

E Dysts benchmark inventory and system definitions

This appendix lists the ten Dysts systems used in the paper-facing $dt\times30$ forecasting benchmark. The systems are drawn from Dysts [11, 12]. We used the continuous-time right-hand sides and metadata from the pinned package version resolved in the project environment, `dysts` 0.96. We use this package under the original license of the repository: Apache-2.0. All ten systems are three-dimensional autonomous flows. The equations below are written in unstandardized native Dysts coordinates (x, y, z); the training cache standardizes coordinates only after trajectories are generated.

Closed-form convention. For each system, the stored observations are generated from $\dot{x} = f_s(x)$ with observation interval $\Delta t = 30\,dt_{\mathrm{native}}$, where dt_{native} is the system-specific Dysts integration interval in Table 5. The learner observes only the resulting stored states, not the vector field, continuous-time solver, or substeps. All ten systems have explicit closed-form right-hand sides in Dysts. Most are polynomial vector fields. The Chua and San–Um–Srisuchinwong systems include absolute-value terms, so they are piecewise smooth rather than globally analytic at the corresponding switching surfaces.

Chua. The Chua system uses the piecewise-linear diode nonlinearity

$$h(x) = m_1 x + \frac{1}{2}(m_0 - m_1)\big(|x + 1| - |x - 1|\big),$$

with parameters

$$\alpha = 15.6, \qquad \beta = 28, \qquad m_0 = -1.142857, \qquad m_1 = -0.71429.$$

The vector field is

$$\begin{aligned}\dot{x} &= \alpha\big(y - x - h(x)\big),\\ \dot{y} &= x - y + z,\\ \dot{z} &= -\beta y.\end{aligned} \tag{20}$$

Dadras. With parameters

$$c = 2, \qquad e = 9, \qquad o = 2.7, \qquad p = 3, \qquad r = 1.7,$$

the Dadras system is

$$\begin{aligned} \dot{x} &= y - px + oyz, \\ \dot{y} &= ry - xz + z, \\ \dot{z} &= cxy - ez. \end{aligned} \tag{21}$$

Dequan Li. With parameters

$$a = 40, \qquad c = 1.833, \qquad d = 0.16, \qquad \varepsilon = 0.65, \qquad f = 20, \qquad k = 55,$$

the Dequan Li system is

$$\begin{aligned} \dot{x} &= a(y - x) + dxz, \\ \dot{y} &= kx + fy - xz, \\ \dot{z} &= cz + xy - \varepsilon x^2. \end{aligned} \tag{22}$$

Hadley. With parameters

$$a = 0.2, \qquad b = 4, \qquad f = 9, \qquad g = 1,$$

the Hadley system is

$$\begin{aligned} \dot{x} &= -y^2 - z^2 - ax + af, \\ \dot{y} &= xy - bxz - y + g, \\ \dot{z} &= bxy + xz - z. \end{aligned} \tag{23}$$

Lu–Chen–Cheng. With parameters

$$a = -10, \qquad b = -4, \qquad c = 18.1,$$

the Lu–Chen–Cheng system is

$$\begin{aligned} \dot{x} &= -\frac{ab}{a+b}x - yz + c, \\ \dot{y} &= ay + xz, \\ \dot{z} &= bz + xy. \end{aligned} \tag{24}$$

Qi–Chen. With parameters

$$a = 38, \qquad b = 2.666, \qquad c = 80,$$

the Qi–Chen system is

$$\begin{aligned} \dot{x} &= a(y - x) + yz, \\ \dot{y} &= cx + y - xz, \\ \dot{z} &= xy - bz. \end{aligned} \tag{25}$$

Sakarya. With parameters

$$a = -1, \qquad b = 1, \qquad c = 1, \qquad h = 1, \qquad p = 1, \qquad q = 0.4, \qquad r = 0.3, \qquad s = 1,$$

the Sakarya system is

$$\begin{aligned} \dot{x} &= ax + hy + syz, \\ \dot{y} &= -by - px + qxz, \\ \dot{z} &= cz - rxy. \end{aligned} \tag{26}$$

San–Um–Srisuchinwong. With parameter $a = 2$, the San–Um–Srisuchinwong system is

$$\begin{aligned} \dot{x} &= y - x, \\ \dot{y} &= -z\tanh(x), \\ \dot{z} &= -a + xy + |y|. \end{aligned} \tag{27}$$

Shimizu–Morioka. With parameters

$$a = 0.85, \qquad b = 0.5,$$

the Shimizu–Morioka system is

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= x - ay - xz, \\ \dot{z} &= -bz + x^2. \end{aligned} \tag{28}$$

Wang–Sun. With parameters

$$a = 0.2, \qquad b = -0.01, \qquad d = -0.4, \qquad e = -1, \qquad f = -1, \qquad q = 1,$$

the Wang–Sun system is

$$\begin{aligned} \dot{x} &= ax + qyz, \\ \dot{y} &= bx + dy - xz, \\ \dot{z} &= ez + fxy. \end{aligned} \tag{29}$$

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